MATH-351

Numerical Methods

Week/Lecture 13

Numerical Methods for solution of differential Equation

Differential Equation

Ordinary differential Eqs. (ODEs)

- First Order ODEs
- Higher Order ODEs

Partial differential Eqs. (PDEs)

Examples

$$\frac{dy}{dx} = e^{x^2}$$

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 4 \quad ; \quad 3\frac{d^3y}{dx^3} + 4y\frac{d^2y}{dx^2} + y = x^2$$

Solution of First Order ODE

$$\frac{dy}{dx} = x \quad \text{(seperable)}$$

$$\int dy = \int x \, dx$$

$$\boxed{y = \frac{x^2}{2} + c}$$

$\rightarrow$ General solution

* Canot approximate function through ~~number~~ numerical Methods.

→ To find C, we need a condition (Initial Condition)

Let $y(0) = 1$, that is

$x = 0 \quad y = 1$ use in general solution

$(0, 1)$

$1 = 0 + c \Rightarrow \boxed{C = 1}$ → put in general solution.

$$\boxed{y = \frac{x^2}{2} + 1}$$ → Particular solution.

- Still we cannot find the particular solution using numerical methods.

** Numerically a solution can be approximated at particular points of domain. That is

Let $0 < x < 4$ with $h = 0.5$

x	y
0	1
0.5	1.125
1	1.5
1.5	2.125
2	3

Numerical Solution first Order ODEs

Given

i) Differential Equation; $\frac{dy}{dx} = f(x, y)$

2) Initial Condition; $y(x_0) = y_0$

3) Domain; $x_0 \leq y \leq x_n$

4) Interval or step-size; h

Methods

i) Euler's Method

2) Improved Euler's Method
Modified Euler's Method
Heun's Method

3) Runge-kutta Methods
(RK Methods)

a) 2nd Order RK Method

b) 4th Order RK Method.

Numerical Solution of 1st Order ODEs

Given 1. Differential Equation

General form $\frac{dy}{dx} = f(x, y)$

2. Initial condition

General form $y(x_0) = y_0$

$x = x_0$; $y = y_0$ (x_0, y_0).

3. Domain

$x_0 \le x \le x_n$ $\cancel{a \le x \le b}$

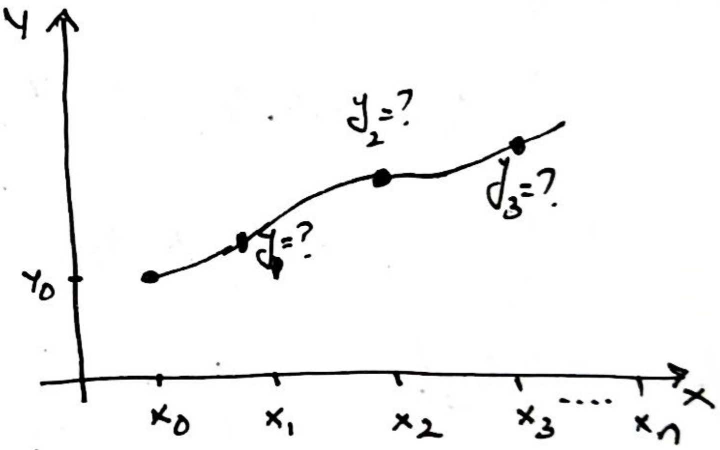
4. Interval or step size $h = \frac{b-a}{n} = \frac{x_n - x_0}{n}$

Graphical Explanation

We have to approximate

$y_1 = ?$ $y_2 = ?$ $y_3 = ?$

------ $y_n = ?$



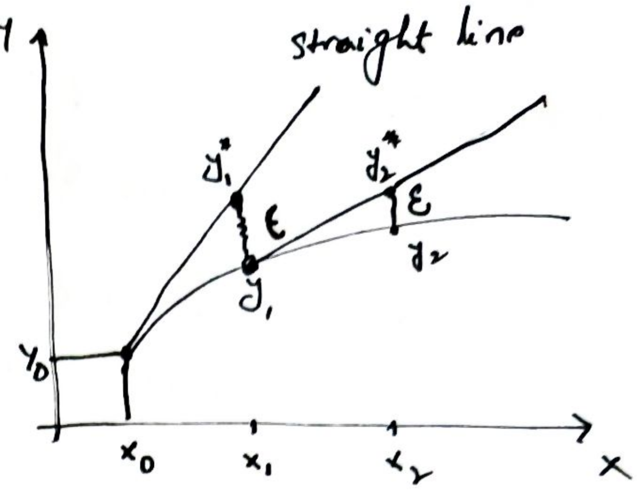
Methods

1) Euler's Method
2) Improved Euler's Method
Modified " "
Heun's Method.
3) 2nd Order RK Method.
4) 4th " " ".

Euler's Method

$$\frac{dy}{dx} = f(x,y)$$

$$y(x_0) = y_0 \quad (x_0, y_0)$$



slope

$$m = \frac{dy}{dx}\bigg|_{(x_0,y_0)} = f(x_0,y_0)$$

$$\boxed{m = f(x_0,y_0)} \rightsquigarrow (1)$$

Eq. of straightline $\quad y - y_1 = m(x - x_1)$

$$\boxed{y - y_0 = f(x_0,y_0)(x - x_0)} \rightsquigarrow (2)$$

when $x = x_1 \quad y = y_1^*$ use in (2)

$$y_1^* - y_0 = f(x_0,y_0)(x_1 - x_0)$$

we know $x_1 - x_0 = h$

$$\boxed{y_1^* = y_0 + h\, f(x_0,y_0)}$$

Euler's Method $\quad \boxed{y_1^* \approx y_1}$

$$\boxed{y_1 = y_0 + h\, f(x_0,y_0)}$$ ✓

Now we have point (x_1, y_1)

slope $\quad m = \frac{dy}{dx}\bigg|_{(x_1,y_1)} = f(x_1,y_1)$

Eq. of straight line through (x_1, y_1)

$$\boxed{y - y_1 = f(x_1,y_1)(x - x_1)}$$

When $x = x_2$ $\quad y = y_2^*$

$$y_2^* - y_1 = f(x_1, y_1)(x_2 - x_1) \qquad x_2 - x_1 = h$$

$$\boxed{y_2^* = y_1 + h\, f(x_1, y_1)}$$

Euler's Method $\quad y_2^* \approx y_2$

$$\boxed{y_2 = y_1 + h\, f(x_1, y_1)}$$

New point is (x_2, y_2)

Via induction

$$\boxed{y_3 = y_2 + h\, f(x_2, y_2)}$$

General form

$$\boxed{y_{n+1} = y_n + h\, f(x_n, y_n)}$$

Formula for Euler's Method.

Example $\quad y' = y - x \qquad y(0) = \frac{1}{2}$

$f(x,y) = y - x \quad ; \quad x_0 = 0 \quad ; \quad y_0 = \frac{1}{2}$

* Use Euler's method with $h = 0.1$ to approximate $y(1)$.

$h = 0.1 \qquad\qquad 0 \leq x \leq 1$

n	x_n	y_n	$f(x_n, y_n)$	$y_{n+1} = y_n + h f(x_n, y_n)$
0	$x_0 = 0$	$y_0 = \frac{1}{2}$	$f(x_0, y_0) = f(0, \frac{1}{2}) = \frac{1}{2}$	$y_1 = y_0 + h f(x_0, y_0) = \frac{1}{2} + (0.1)(\frac{1}{2}) = 0.55$
1	$x_1 = 0.1$	$y_1 = 0.55$	$f(x_1, y_1) = f(0.1, 0.55) = 0.45$	$y_2 = y_1 + h f(x_1, y_1) = 0.55 + (0.1)(0.45) = 0.595$
2	$x_2 = 0.2$	$y_2 = 0.595$	$f(x_2, y_2) = f(0.2, 0.595)$	$y_3 = y_2 + h f(x_2, y_2) = 0.6345$
				$y_4 = 0.66795$
3	$x_3 = 0.3$			$y_5 = 0.694745$
4	$x_4 = 0.4$			$y_6 = 0.714219$
	$x_5 = 0.5$			$y_7 = 0.725641$
	$x_6 = 0.6$			$y_8 = 0.728205$
	$x_7 = 0.7$			$y_9 = 0.721026$
	$x_8 = 0.8$			$y_{10} = 0.703129$
	$x_9 = 0.9$			
	$x_{10} = 1.0$	$y_{10} = 0.703129$		